Data Science with Python
Ticket Out the Door
Set 28 Ticket Out The Door

Name ____ Period ______

Skill 28.1 Exercise 1

Refer to the *titanic* dataframe below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

In the space below, make crosstab table that compares *Sex* to *Survived*

```
crosstab_counts = pd.crosstab(titanic['Sex'], titanic['Survived'])
```

Convert the frequencies in the crosstab table below to proportions.

```
crosstab_proportions = crosstab_counts/len(titanic)
```

The actual crosstab chart which compares Sex and Survived as proportions is shown below.

Survived	**0**	**1**
Sex		
female	0.090909	0.261504
male	0.525253	0.122334

Based on the proportions in the table, does survival appear to be associated with sex? Justify your answer using specific values.

Yes, survival appears to be associated with sex. According to the table, approximately 26.15% of all passengers were females who survived, while only about 12.23% were males who survived. In contrast, 52.53% of all passengers were males who did not survive, compared to only 9.09% who were females who did not survive. These differences suggest that females were much more likely to survive than males, indicating an association between sex and survival.

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Skill 28.2 Exercise 1

In the previous exercise, we calculated the proportions for the survivors and sex.

Survived	0	1
Sex		
female	0.090909	0.261504
male	0.525253	0.122334

Calculate the marginal proportions below (row totals and column totals)

Survived	0	1	Row totals
Sex			
female	0.090909	0.261504	.352413
male	0.525253	0.122334	.647587
Column totals	.616162	.383838	

Calculate the expected proportions,

Survived	0	1
Sex		
female	.217	.125
male	.399	.249

Calculate the expected frequencies by multiplying proportion by the sample size (891 for this data)

Survived	0	1
Sex		
female	193	121
male	356	222

What does the table above show?

The table above tells us that if there were no association between the sex and survivors, we would expect 193 females to have died, 111 females survived, 356 males to have died, and 222 males to have survived.

The actual frequencies are shown below,

Survived	0	1
Sex		
female	81	233
male	468	109

Compare the calculated frequencies to the actual frequencies. What do these tables tell us about the association between these two variables?

The differences between the calculated and actual tell us that sex and survival are indeed associated

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Skill 28.2 Exercise 2
Use the *chi2_contingency* function to calculate the expected counts and print the results
`chi2, p, dof, expected = chi2_contingency(crosstab_counts)` `display(expected)`

Skill 28.3 Exercise 1
Refer to the *chi2_frequency* function you applied in the previous example. Print the *chi2* statistic.
`print(chi2)`
Fill in the blank below.
For a 2x2 table (like the one we've been investigating), a Chi-Square statistic larger than around ______ would strongly suggest an association between the variables.
The actual *chi2* statistic is 261. What does this value tell us about the association of the *sex* and *survivor* variables?
The Chi-Square statistic is much larger than 4! This adds to our evidence that the variables are highly associated.